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Article

The adversary: the philosophy of cryptography

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Abstract

The essence of cryptography is secure communication in the presence of adversaries. We present an analysis of cryptography in terms of a philosophy founded on Dretske's classical philosophy of information, which naturally is built on the work of Shannon and is so amendable to being extended with the notion of adversary. The attempts to transform cryptography from an art to a science were based on formalizing the adversary, but these very formalizations fall short in the face of real-world adversaries, as demonstrated by the Snowden leaks. Nevertheless, this problem is not unique to cryptography *per se*, but a problem inherent in the process of abstraction itself, and so we sketch a research agenda wherein overcoming the limits of abstraction necessitates the embrace of complexity and empirical data, as well as the questioning of our own ontological assumptions.

Keywords: philosophy; cryptography; secrecy; history

Introduction

The internet's promise of universally connecting the world has resulted in a seemingly infinite space teeming with new and unknown adversaries, and cryptography is one of the few crucial mechanisms of defense possible in this new world. Yet, while there has been much ado about information theory in philosophy [1,2], there has been little philosophical work on cryptography despite cryptography itself being the origin of information theory [3]. In order to bring cryptography to the attention of philosophy, as well as elucidate the possible contributions of philosophy to cryptography, we outline the crucial concepts of cryptography from a historical perspective, while in turn applying philosophical critique to these concepts. Based on this analysis, our contribution is then to sketch out a research agenda for the broader field of cybersecurity that benefits from the insights and tools given by philosophy.

For most of human history, issues of security have involved physical dangers posed by relatively known groups of adversaries, and trust confined to small bands of humans, such as the family and the tribe. In the last century, the development of large-scale social institutions, such as nation-states, has grounded trust between unknown parties in explicit social norms and bureaucratic procedures at the cost of surveillance of potential adversaries, including its own population, while its own information organization becomes increasingly dependent on secrecy [4]. Cryptography, the study of techniques for secure communication in the presence of adversaries, makes the revolutionary and ultimately philosophical claim that secrets can be communicated in the digital world via the application of mathematics. It is precisely this claim that we will historicize and philosophically interrogate, as it leads to the possibility of new notions of trust and security if adversaries can be bound by the chains of cryptography.

First, we discuss the intuitive concept of a *secret* and how it fits within the philosophy of information as initially theorized by Dretske [1]. Then we turn to how the problem of the preservation of secrets in the presence of adversaries led to Shannon's first formalization of cryptography [3]. Next, we turn to the historical development of cryptography and the debates, which were ultimately philosophical in nature, over its increasing abstraction, and inspect the theoretical limits of these models. Surprisingly, we then show how an entirely different type of adversary presented itself: the global mass surveillance system of the National Security Agency (NSA) of the United States of America as revealed by Edward Snowden, effectively bringing the field of cryptography to a crisis point by revealing the practical limits of these formal models. By returning to our initial philosophical analysis in the conclusion, we propose that a paradigm based on complex networks with real-world data is needed to properly philosophically analyse the wider socio-technical system of adversaries and secrets.

In summary, we offer more of a broad-ranging philosophical inquiry into secrecy in the digital era than a rigorous formal theory of secrecy. Nonetheless, our goal is to lay the groundwork for a more empirical and model-based critique of the limits of existing formalizations. This method of construction via critique should be one

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proper role of the philosophy of any science, including the philosophy of cryptography, even if there is not agreement on the conclusions.

Philosophical preliminaries

In order to begin any philosophical analysis of cryptography, a preliminary definition of terms is needed. Although our goal is to provide insight into the philosophical nature of the adversary rather than create philosophical adversaries, we will nonetheless have to make a number of perhaps problematic philosophical assumptions at the outset. Each of these assumptions can provoke whole debates in philosophical fields from metaphysics to epistemology. On those debates, while it would best to be rigorously neutral, we will nonetheless have to sketch a minimal picture in order to get our inquiry off the ground. In doing so, we will draw on the classical analysis of Fred Dretske in *Knowledge and the Flow of Information* [1], while reverting to the minimalist metaphysical assumptions and ontological pluralism of Brian Cantwell Smith's *On the Origin of Objects* [5] in order to deploy Cantwell Smith's critique of abstraction to cryptography.

To begin with a brief foray into metaphysics, the world can be considered a primordial metaphysical flux from which local states of affairs arise. These states of affairs may be relatively *proximal*, and so capable of mechanistic causal interaction where they affect each other necessarily, or *distal*, such that these states of affairs cannot affect each other [5]. The world can thus be "cut" at various joints when analysed in terms of separate systems, although the boundaries are often dynamic over time at best and arbitrary at worst. Following a traditional reading of Dretske, *information* consists of regularities in these states of affairs that may be transmitted and so lead to interaction between these systems [1].

Information can be transmitted via various information-carrying vehicles, often called *messages* or *signals*, between systems, which may include agents who have the capability of perceiving information through various sensory and perceptual apparatuses. Via their autonomous agency, *agents* take actions upon the world and possess a proximal *local state* that they can access to determine their actions. Agents such as humans can be distinguished from mere information-carrying vehicles such as DNA and thermometers by virtue of having commitments to their own intentional structure: thoughts, plans, beliefs about the world, a commitment to a world *as* a world, and the ability to change the world via their own agency [5]. However tenuous and changing it may be, there is a boundary between themselves and the world, and between themselves and other distal agents.

Agents also may internalize information and so can be thought to "posses" or, informally, to "know" the information via incorporating information into their local state, including via cognition. Rather than be led astray in epistemological debates, it is enough to say that agents then have *access* to information when they can incorporate it into their local state. Agents such as humans can communicate information to other agents via encoding information as an information-bearing *message* in an analogue or digital manner [6]. Agents then communicate with each other to share information and coordinate behavior. This leads to the phenomenon of distributed cognition, where agents coordinate their cognition via the "propagation of representational state across a series of representational media" [7]. This serves as the foundation of *information organization*, where information is shared between various collective groups of agents, and others are excluded. These excluded agents may nonetheless want access to this information, and are so thought of as adversaries, a key notion to which we will return.

The end result of these preliminaries is that we have the language to discuss the nature of a secret philosophically, while still grounding ourselves on the traditional paradigm of Shannon information theory in which information is sent as messages through a channel between a sender and a recipient, which themselves are agents [8]. With these preliminaries out of the way, let us get to the heart of the philosophical matter: What is a secret? Informally, a secret is information not known by everyone, but only by a few, who can purposefully exclude this information from access by adversaries. In other words, a secret is information possessed by some agents, but not all, who try to limit the flow of this information.

One can further define the nature of a secret by two conditions. First, a secret is a relationship between one or more agents, called *secret keepers*, and the information. Second, a secret is relative to the adversaries that desire to possess the secret, where any possible adversaries are excluded from access to the information by the secret keepers. Thus, secrecy is a relationship over information between secret keepers and adversaries, where secret keepers are defending their secrets from various adversaries, and the secret keepers and adversaries should be disjoint sets of agents as long as some particular information is to remain a secret.

Any such information itself is not intrinsically a secret *per se*; what is a secret to one group may not be a secret to another group. For example, a map—even if it has the location of underground American nuclear warheads in Europe—is not in and of itself a secret, but only a secret in relation to a set of secret keepers, such as the US intelligence agencies who do not want this map to be more widely known to their adversaries like Russia. For a secret to remain a secret, it must not be transmitted to just anyone, but only to the intended recipient(s) who maintain its secrecy and so join the ranks of the secret keepers. However, for the sake of brevity, the concept of "secret information relative to a group of secret keepers" will be abbreviated simply as "secret."

This brings the question of the *adversary* to the forefront. The *adversary* is an agent that is trying to gain access to secret information despite not being a secret keeper. When the adversary gains access to this information, the secret is *revealed* and the information is no longer a secret. The secret only has to be revealed by the adversary to themselves, not the general public, and the very fact that the secret has been revealed may be kept secret by the adversary themselves. For secrecy to be true, no adversary can possess the secret. The secret keepers and the adversary are thus in a state of eternal opposition, where the secret keepers try to defend their secrets from the adversary. The secret keepers may know the adversary quite well: for example, the adversary of the American agency in terms of nuclear weaponry was the Russian intelligence agency, and *vice versa*. However, the secret keepers may not know the adversary is trying to reveal their secrets; the secret keepers may consider them harmless or even an ally. The Ethiopians originally assumed the Italians were friendly until a translator revealed that the Italians were trying to gain access to Ethiopian secrets in order to colonize their country [9]. The secret keepers may not even be aware of the success of their adversary in revealing their secrets; the Third Reich did not know the Enigma machines were cracked by British intelligence at Bletchley Park [10]. In the extreme case, the secret keepers do not have to know that an adversary exists: It is increasingly common that an unknown anonymous group may hack into a group of computers and that secrets may be revealed, but to whom is a matter of mystery. Concretely, an adversary could corrupt a secret keeper's device to access the message, but they may also passively record all information sent across a channel, which is usually imperceptible to their targets. For this reason, secrets thus are typically protected against unknown adversaries.

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No secrets are eternal; they hold only for a period of time. So no information is a secret in of itself. A secret involves a relative relationship between information and secret keepers over a period of time relative to the adversaries and the wider world, such that no secret can be absolute. Any secret information in a message must be actively kept secret for the relationship of secrecy to continue; the behavior and information organization of the secret keepers is structured to protect their secret. For example, take the paradigmatic category of classified information, such as US State Department cables. Once they were released into the world by Wikileaks, anyone could access them and read the information: the cables were no longer *de facto* secrets even if the cables were secrets *de jure*. These cables were only a secret until they were leaked. Revealing a secret does not mean that everyone knows the information (as in a leak), but merely that the information is known outside the group of secret keepers.

Once a secret is revealed, it may be kept as a secret by the adversary themselves or it may be released to the general public via a *leak*, and so the secret becomes public knowledge. It is useful to distinguish public knowledge from common knowledge. Lewis defines "common knowledge" as the kind of information in a population of agents, wherein everyone in the population has reason to believe that a particular state of affairs holds [11]. What we need is a weaker notion of *public knowledge*, which is information that any agent could potentially have access to, but does not necessarily know (as is the case with common knowledge). A secret is the opposite of public knowledge and so "uncommon knowledge." Interestingly enough, one of the ramifications of this common-sense view of secrecy is that the opposite of secrecy is *publicity*, such that what is public cannot be secret.

Heavyweight epistemological terms are bandied about informally in cryptography, such that an agent like an adversary "knows the secret" or secret keepers "believe the secret is secure." Although the tricky epistemological question of belief is part and parcel of secrecy, a proper treatment of secrecy would be book-length, and so it should suffice to note that these terms are used informally to designate a relationship of access rather than justification or even belief. For example, secret keepers must believe that the information is secret so that they work towards keeping it a secret. Does this mean that they believe or even know the secret itself? Take the example of the first known use of steganography in which Histiaeus of Miletus tattooed a message warning of a Persian attack on his poor slave's scalp in a language the slave likely could not read. Was the slave a secret keeper? In more modern times, an entire apparatus of agencies exist to keep secrets such as maps of US warheads, but not every member knows the information in the map. In our analysis, secret keepers do not have to personally know the secret (although they may), and so secret keeping typically includes a large number of agents who do not know the secret but nonetheless work to preserve knowledge of the secret from adversaries. Furthermore, secret keepers themselves may have access to the information that comprises the secret but may not believe in the truth of the secret, such as intelligence of uncertain or unreliable origin, and such information is often kept secret just in case it is true. However, secrecy is not a relationship of mere belief, for if a secret is revealed by an adversary, it is no longer secret even if the secret keepers believe the relationship of secrecy still holds.

While working cryptographers may consider this all an unnecessary metaphysical and ontological excursion, the point of this section is to familiarize philosophers with cryptography by showing how cryptographic concepts, once grafted on top of relatively well-known philosophical theories of information, can remain neutral to questions of epistemology by thinking in terms of access to information rather than belief or knowledge [1]. In essence, secrecy is relative to the adversary, as the adversary is excluded from access to the information by the secret keepers. The adversary is trying to reveal the secret and, in the extreme case, even make the secret public via a leak. The information organization of secret keepers expands beyond those with personal knowledge of the secret, to a set of techniques and behaviors to defend the secret against possibly even unknown adversaries. The philosophy of cryptography can then be thought of as a progression of conceptual models of the adversary throughout history. It is to this question that we turn next.

Foundations of modern cryptography

Cryptography is the study of how to communicate secrets without revealing these secrets to an adversary. The origin of the term cryptography comes from the Greek *kryptos*, meaning "hidden," and *graph*, which means "writing," so that cryptography is literally the "art of secret writing." So far, we have not discussed the precise techniques used by secret keepers to hide secrets from adversaries, and before the advent of modern cryptography these cryptographic techniques were varied and not subsumed theoretically in a single paradigm [10]. The crucial breakthrough was the creation of the first formal model of the adversary, where its powers in leaking secret information could be mathematically analyzed. In a move that has confused generations of philosophers, this required stripping away questions of semantics from information, and analyzing information purely in terms of its encoding as done by Shannon [8]. The motivation for this theoretical revolution that led to information theory was Shannon's cryptographic work, which was classified at the time [3].

Shannon's formalization of cryptography

Modern cryptography can be dated to the publication of "Communication Theory of Secrecy Systems" in 1949 in which the terms of cryptography were for the first time given a general mathematical treatment, with proofs provided for security in terms of probability theory [12]. As Shannon outlines in his "A Mathematical Theory of Communication" [8], the creation of modern cryptography appears to be a predictable result of prior published work in defining the communication of information *in general* over a channel between a sender and a receiver. The only addition to classical Shannon information theory is that some transformation should be applied—namely encryption—that makes the message unreadable to the adversary by transformation into a *ciphertext*, while the intended recipient can equally apply another transformation—decryption—to reveal the message. Given that Shannon's model of information is also deployed philosophically by Dretske [1], the philosophical preliminaries in Section 2 neatly map to a conceptual analysis of modern cryptography.

There is a small wrinkle in traditional philosophical accounts of information and cryptography; rather than building cryptography in terms of information theory, it should be noted that Shannon worked in Bell Labs on classified research for the USA involving cryptography during the Second World War, and his original publication on the formal foundations of cryptography, entitled *A Mathematical Theory of Cryptography* was actually written years before his more lauded publication on information theory called *A Mathematical Theory of Communication* [3]. Therefore, the fundamentals of Shannon information theory, including such notions as entropy, were actually developed for cryptographic systems and not for general purpose telephone communication as is commonly believed. It has little been explored that cryptography is the *central*

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motivation of Shannon information theory, even though philosophers like Bar-Hillel, Carnap, Dretske, and more recently Floridi have found Shannon to be central to their philosophical work [1,2,13].

To rephrase the general problem, we assume there are two agents—a sender and a receiver—communicating secret information encoded into a number of bits. This message must be given to the sender in some language "in the clear" (i.e. *cleartext*, which can be understood), but the message needs to be sent through a *cipher* to become ciphertext (or a "cryptogram") that an adversary cannot understand, even if intercepted. A system of communication between secret keepers is considered secure if the adversary cannot reveal the secrets. For Shannon, the system should not depend on the meaning—the semantics—of the information, but only on how it is encoded. The process of transforming the encoding of a message into ciphertext is *encryption* and its reverse is *decryption*, where the method of encryption and decryption combined can be considered the *cryptosystem*. The cipher is assumed to be some mechanical system, such as an algorithm implemented on a computer, but—as Shannon's maxim states—it should be assumed that "the enemy knows the system" [12]. While the design of the system should be public, there is assumed to be a part of the system that is a secret from the adversary: The *key* used to encrypt and decrypt, and the security of the entire system should be reducible to the secrecy of the key, the ancient principle of Kerckhoffs' principle [14]. The adversary is assumed to be able to commit to *cryptanalysis*, which means being able to use techniques such as statistical analysis to attempt to reveal the secret without having access to the secret key. Shannon assumes that the sender and receiver both have secret keys.

Shannon's first controversial, yet revolutionary, move is to get rid of any concept of the meaning of the information, as he is assuming the information is encrypted in such a way as to hide its meaning on the level of the bit-wise encoding, and he assumes if the message was decrypted then the secret would be revealed and fully semantically transparent. These are plausible assumptions given the nature of wartime cryptography, although it is less clear if these assumptions are plausible for a full-fledged philosophical analysis of information [2,6]. Using this framework, Shannon proves two surprising results that shape to this day the landscape of modern cryptography. First, Shannon defines *theoretical secrecy* (often called "Shannon security" or "informationally theoretically secure") as: "How immune is a system to cryptanalysis when the cryptanalyst has unlimited time and manpower available for the analysis of cryptograms?" In other words, a system is theoretically secret—perfectly secret—if there is a way to encrypt a message so that no amount of computation can break it to reveal the secret [3]. Second, Shannon then proves that in order for a system to be perfectly secret, the system must have a key of equal length to the message itself (i.e. a "one time pad"), which means that "perfect secrecy requires an infinite amount of key if we allow messages of unlimited length." [3]. To the eternal disappointment of any simple-minded approach to encryption, Shannon then proves that if this key is reused, the cryptanalyst can in theory break the encryption.

While perfection may be impossible, there is a ray of hope, as Shannon proposes a new definition of *practical secrecy*: The amount of "work" that is "characteristic of the system [...] taken over all messages and all keys with their appropriate probabilities," where the work can be measured in the amount of computation it takes to decrypt an encrypted message [12]. Thus, the idea is that the adversary must have a finite amount of time and power to break the decryption, and so keys shorter than the message must be possible, but the exact cryptosystem to be used was not given and the powers of the adversary were not formalized by Shannon. Due to the classified nature of Shannon's work and cryptography in general, work in cryptography disappeared in the USA into the National Security Agency, formed in 1952 and superseding the peacetime cryptanalysis done by Black Chamber and during the Second World War by the Signal Intelligence Service.

To take a step back, what is truly underappreciated by both historians and philosophers is that Shannon wrote his *A Mathematical Theory of Cryptography* in 1945 before he published his much more famous *A Mathematical Theory of Communication Systems* in 1948. In fact, Shannon's classical information theory is simply his theory of the communication of secrets purged of any cryptographic elements and is not widely taught or even known outside cryptography. Due to his original work being classified during the Second World War, Shannon was only able to publish his declassified wartime work as the *Communication Theory of Secrecy Systems* in 1949 after his information theory gained popularity. In this manner, the entire concept of information is a by-product of cryptography, as is the invention of modern computers [10], a historical fact lost upon philosophers who tend to focus on cybernetics and artificial intelligence. As philosophers like Dretske based their work on Shannon information theory [1], Shannon's formalization of the power of the adversary in terms of perfect and practical secrecy fits neatly on top of our initial philosophical analysis of information.

Access control and information organization

Yet even though cryptography disappeared from the public view to become the province of intelligence agencies, governments still needed control to avoid revealing secrets. The task of keeping secrets between secret keepers can be thought of as *access control*, where there is information and a *policy* that determines which agents may have access to the secrets. While the secrecy of particular information-bearing messages may require cryptography, access control is the foundation of the information organization of secrets in general. For example, there have been attempts to use one-time pads to create perfectly secure secret-sharing systems, and for a long-period of time Soviet communications required the use of one-time pads. Yet, this kind of cryptosystem under Shannon's paradigm had a fatal flaw: each message required a new key. This means that the Soviet encryption system essentially had to centralize the production of key material in Moscow so that all secret communications went through a single point in Moscow, even for covert communications between two agents inside the USA. The sheer hassle of the amount of key material being produced led to accidental key reusage, which in turn enabled American intelligence to successfully decrypt messages [15]. It is not just the techniques of cryptography but the social system of the secret keepers that is required to keep a secret. Adversaries are not limited to technical means, they may actively be attempting to reveal secrets of their targets by intercepting the message, but also monitoring or interfering with—or even subverting—the behavior of secret keepers and their information organization.

The origins of access control systems typically come from the military, where secrets must be strictly restricted to secret keepers, where there is a hierarchical degree of "levels" of classification ranging in the US government from "top secret," merely "confidential," to completely "unclassified"—although additional qualifications such as "NOFORN" (no foreign national) complicate the seemingly simple hierarchy into a more "lattice"-like structure [16]. These policies originated as human-centric levels of access control. However, the spread of digital computing sparked concerns that foreign governments and other adversaries would attack government computer systems and networks to gain access to secrets. It thus became

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imperative that this line of research get transferred to computer systems [17]. In particular, the deployment of the Multics operating system, where many users shared a single computer and so could easily share information, began to highlight the problem of maintaining the confidentiality of easily copied and digitally encoded information [18]. Formalized models of these levels of access control, in order to attach them to computing systems, became an area of active work for the military and intelligence agencies, with concepts and techniques slowly drifting into an emerging and more public field of *security engineering* [19].

Access control systems with a security model were first formalized in the Bell–Padula model, which effectively aimed to keep secrets between groups of secret-keepers arranged in a strict hierarchy, and where agents on the "top" of hierarchy could transfer information between themselves and read information of those lower in the hierarchy [20]. Another issue was being certain messages maintained their integrity, i.e. that agents at a lower-level could not alter information meant for a higher-level [21]. The more complex lattice-structure used by the US government and inside many organizations, where different agents can have different levels of access control, ended up being formalized much later [16]. In contrast to cryptography, these access control systems are typically enforced via the computer's administrator or operating system that creates the various groups and user identities. One of their most ambitious research agendas was to create, in contrast to Multics, a provably secure operating system [22].

Three important security properties are typically desired by early access control systems, which were then later taken to be requirements for cryptosystems. The first and most important is *confidentiality* to ensure that a secret is only revealed to its intended recipient(s), thereby keeping the secret in the circle of secret keepers. There are two more properties that can be accomplished by cryptography that do not on the surface have anything to do with confidentiality. The second is *integrity*, namely that the information encoded in a message has not been changed en route over a channel. The third property is *authentication*, such that a recipient can verify that it was indeed the sender that sent the message. An additional fourth property is often implied by authentication, namely that of *nonrepudiation*, so that a sender cannot deny having sent a message to a recipient. These latter two properties of integrity and authentication may seem divorced from confidentiality, but in fact are crucially important to maintaining secrecy. The property of authentication is required to identify secret keepers and so separate them from the general population and possible adversaries, and integrity is then required to ensure that the secret itself is maintained, the information contained in the message is not accidental or intentionally lost. Typically it should be noted that in the wider security community, the term "availability"—information is available when required—was often substituted for authentication, although availability is not a security property.

Beyond Shannon's analysis of cryptography, what is also not properly appreciated is the invention of access control modeling as a science of information for the organization of secret keepers, as it was mostly hidden from the public and only used by governments as necessitated by digital computing. However, early access control models assume a centralized information organization with the ability to identify users and distribute cryptographic secret keys. When used in the real world, as shown by the centralization of one-time pads by Russian intelligence in Moscow, which eventually led to its failure due to accidental key reuse [23], these systems quickly became cumbersome and difficult to deploy in a ever-changing dynamic world of secrets and adversaries.

Formalizing the adversary

Two philosophical questions bedevilled the use of cryptography. The first was how cryptography could be used by secret keepers that were not part of a centralized organization that could deliver secret keys. The second question was how could it be guaranteed that a real-world adversary, as opposed to the perfect adversary imagined by Shannon, could not break a particular cryptographic encoding of a message by secret keepers.

The revolution of public key cryptography

Up until 1976, cryptosystems were *symmetric* key-based systems of the type formalized by Shannon, where the sender and receiver used a secret key for both decryption and encryption. Yet for messages to be encrypted and decrypted by a secret key, both the receiver and the sender had to have their secret key somehow securely given to them via some secure channel. In other words, it was generally assumed that the keys were distributed by a trusted third-party in the larger organization of secret keepers. A trusted third-party is an element of a communication protocol that is trusted not to reveal any secret. In practice, this tended to limit the use of cryptography in practice to centralized social organizations such as governments with necessary infrastructures. This, and the fact that most cryptographers were employed by intelligence agencies, meant that the public academic field of cryptography barely existed, although the publication of Kahn's *The Codebreakers* in 1967 renewed interest in the development of a possible science of secrets [10]. Academics such as Whitfield Diffie and Martin Hellman began to question whether or not the usage of cryptography was possible over insecure channels by multiple users when observed by an adversary without preexisting secret keys [24].[1]

The invention of public-key cryptography by Diffie and Hellman in their *New Directions in Cryptography* [25] solved the problem of key distribution amongst an open-ended number of parties over insecure channels via the creation of *public key cryptography*. Their central insight was to "split" the key into a public key and private key, where two users could then exchange public keys and, via the use of their private key, create a new shared secret key to secure the channel even with the adversary intercepting the public keys and viewing all ciphertext messages. This would allow any person, even those without organizations, to become a secret keeper and use cryptography to protect their secrets from adversaries. The philosophical and even political ramifications of the historic shift in the landscape of sovereignty caused by this breakthrough have yet to be fully appreciated [4].

Earlier, Shannon enigmatically ended his original essay by noting that "the problem of good cipher design is essentially one of finding difficult problems" [12]. Building on this insight, Diffie and Hellman discovered that certain mathematical functions had "trap door" information that allowed an agent with this secret information (such as a key) to reverse a function. Without such information it was impossible, however, and these "one way" functions could be used to build new types of public key cryptosystems, where the private key was the "trap door." As Jacques Stern, the founding figure of French cryptography, put it: "from an epistemological perspective, one can say that Diffie and Hellman have drawn the most extreme consequence" from Kerckhoffs' principle so that the encrypting key can be public and only the decrypting key must be secret [26]. The public

1 A technique for this "nonsecret" cryptography (as the initial keys were not needed to be secret) to do so had in fact been discovered by GCHQ over a decade earlier in 1967, but had remained classified until until 1997 [23].

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key could then just be an exponentiation of the large random number of the secret key. This led to the Diffie–Hellman key exchange protocol in which the public key could be sent over the clear and shown to the adversary, but only then combined with the secret key of the receiver in order to create a symmetric key [25].

Diffie–Hellman key exchange neatly solved the problem of key distribution without a trusted third party, but it failed to actually provide algorithms to encrypt and decrypt messages using public key cryptography. While Diffie and Hellman did list a number of attacks—such as the adversary being able to send arbitrary plaintexts to encrypt (the chosen plaintext attack), or the adversary possessing the decrypted plaintext for arbitrary ciphertexts (the chosen ciphertext attack)—their ground-breaking work also did not provide a method for determining whether an encryption algorithm was secure against unknown adversaries [25]. Whether or not a cryptosystem was secure was hard to judge for proposed cryptosystems, as some problems that seemed to be hard "trapdoor" type problems for worst-case scenarios, such as Merkle's knapsacks [27], would fail against actual adversaries who possessed "average-case" computational power in the real world [28]. Therefore, the field of cryptography simply had to imagine what could be done against known attacks by adversaries. After all, there is no reason to believe that adversaries would limit themselves to known attacks.

Diffie and Hellman stated that "we stand today on the brink of a revolution in cryptography," and the revolution they began was completed by the invention of the Rivest–Shamir–Adleman (RSA) cryptosystem that could encrypt, decrypt, and also sign and verify digital signatures using the factorization of the product of two large prime numbers as the trapdoor function [29]. Further work by Hellman's student El-Gamal then extended the work of Diffie–Hellman Key Exchange to provide encryption and decryption-based on modular exponentiation. Yet, even these cryptosystems effectively were not proven to be secure using mathematical proofs. Attempts to construct a RSA cryptosystem based on standard assumptions, such as the provably secure Rabin cryptosystem [30], discovered that the very properties that made it provably secure could be provably broken by chosen ciphertext attacks [31]. These problems could be solved by the introduction of randomness into the decryption and encryption operations themselves. This would include not just the keys, but also randomness appended to the message before encryption [32]. Yet the exact key sizes, padding, and other matters appeared to be more an art than a science.

Furthermore, real deployed cryptosystems needed to have these parameters set in order to interoperate. After at first attempting to suppress the spread of public-key cryptography, the NSA eventually began to work with National Institute of Standards and Technology (NIST) to standardize cryptosystems for use outside governments. This led to the distinction between cryptographic *primitives*, the components of a cryptosystem whose function is assumed to provide a security property, against cryptographic *protocols* that are constructed out of cryptographic primitives to achieve a more complex function in a secure manner. Examples of cryptographic primitives include hash functions used to create cryptographic one-way "digests" (similar to a unique fingerprint for an encoding of information), ranging from Secure Hashing Algorithm (SHA) to symmetric encryption algorithms like Advanced Encryption Standard (AES), as well as public key systems for encryption, decryption, and digital signatures. As cryptography spread, the standardization of these primitives was led by nation-state level standards bodies that were closely connected to their respective intelligence agencies in terms of cryptography, such as NIST (USA), CRYPTREC (Japan), GOST (Russia), and ETSI (Europe), with standards from the USA dominating deployments. In the early days of cryptographic systems, nation standards were done in an opaque manner, as exemplified by the production of Digital Encryption Standard (DES) by IBM with NSA approval and Digital Signature Algorithm (DSA), an algorithm that was purposefully crippled to prevent confidentiality (which the NSA viewed as dangerous at the time to spread to the public, though it allowed authentication). The USA moved to running open contests like NIST, where cryptographers attacked each other's proposals for primitives. These were produced via cryptographic contests ran by NIST that nonetheless left the question open: How do we know the power of the adversary, and can we prevent unknown attacks?

Formalizing the adversary via computational complexity

The philosophical answer was to formalize the adversary on a level of abstraction so that all possible attacks could be precisely characterized in terms of the theory of computation, and security could be proven against such an abstract adversary. The advent of provable security was began by Shafi Goldwasser and Silvio Micali's precise mathematical formalization of the adversary in terms of computational complexity theory, which could characterize an adversary with a bounded amount of computation as a particular kind of Turing machine operating over binary encoded messages [33]. One way for the adversary to break any cryptosystem is simply to enumerate all possible keys, which on a classical (super)computer still takes an exponential amount of time, and so would not terminate in thousands of years. Thus, an exponential–time algorithm to break a cryptosystem is as good as not having an algorithm for the purposes of practical security. In order to formalize this insight, the adversary can be given a Turing machine that runs in polynomial time; its running time is upper bounded by a polynomial expression in the size of the input, where the input may be any observable information given by the sending of the encrypted message between the sender and the receiver [34]. The adversary can also be considered to have access to uniformly chosen and independent randomness, and so is a probabilistic polynomial–time Turing machine (PPT). To generalize, for each possible input size, the adversary could possess a different algorithm, so the adversary is a nonuniform PPT. A *computationally secure* cryptosystem requires proving that a cryptosystem could be reduced to the computational intractability of a mathematical problem. The proof typically produced bounds on the success of the nonuniform PPT adversary, with a system being secure if the probability of success was shown to be negligible within a bound.

Via this breakthrough, previously informally characterized attacks could be reformulated mathematically, and so systems could be proven to be secure in terms of cryptographic properties like confidentiality. For example, chosen plaintext attacks could be rephrased mathematically via *semantic security* in terms of the polynomial–time (efficient) adversary: "Whatever is efficiently computable about the cleartext given the ciphertext, is also efficiently computable without the ciphertext" [34]. While some probabilistic advantage of the adversary must exist because the plaintext is mixed into the ciphertext by the key, it is bounded to be negligible. This notion could be formalized into the formal property of *indistinguishability* in the face of plaintext and ciphertext attacks, with the eventual result that a particular cryptosystem could be provably secure against these kinds of attacks.

Cryptography started to be systematized in a new Hilbert program for secrecy, becoming as respectable as any other branch of mathematics, a feat made possible through replacing paranoid

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intuitions with provable results from computational complexity. Cryptographers would use "games" to define reductions to known hard problems in mathematics; showing that an adversary must solve a hard problem in order to break the system. As long as the hard problem remained unsolvable by the computationally defined adversary, the cryptographic edifice built upon the hard problem would be secure. There was a veritable golden age in provable security, pushing the boundaries of what was even possible with cryptography, as exemplified by zero-knowledge proofs, where one could prove knowledge of information without revealing anything besides the possession of the knowledge [35]. There were also practical ramifications. For example, the difficulty of breaking a cryptosystem to reveal a secret increases faster than any polynomial function of the key, so the cryptosystem becomes more difficult to break by any polynomial–time adversary simply by increasing the key length.

Could cryptography transform from a mathematical discipline to a scientific method with real-world impact? Provable security was theorized to have potentially huge ramifications for the practical standardization of protocols, which had started with parameters being determined in an *ad hoc* manner. For example, RSA needed common encoding and padding to be used in practice securely, so new standards like Public Key Cryptography Standards (PKCS) #1 were drafted [36]. Standards could depend on *practice-oriented provable security*, where the security properties of various deployed standards could be studied rigorously and proven [37]. Yet with a few exceptions, such as the Cramer–Schoup cryptosystem [38], most cryptosystems—even RSA—could not be proven secure within a standard computational model [39].

The core of the problem was that many real-world primitives, such as hash functions, were not clearly definable mathematically under the standard model of cryptography. Therefore, there were new extensions to provable security that substituted the ideal functionality in proofs for the use of a standard primitive in the real world. For example, the SHA1 hash function could be modeled as a "random oracle model" in order to be used in proofs, where the random oracle maps each possible input to a random output whose infinite output could then be truncated [40]. Likewise, proofs could be made in the generic group model, where an encoding that was efficient (such as the finite fields used by RSA) could be substituted with a random encoding [41]. Trusted third-parties whose behavior was considered nonadversarial could also be invoked for the need in proofs, such as the setup of a common reference string in zero-knowledge proofs [42].

Progress in formalization led to a veritable blossoming of a thousand proofs, with proofs for useful new constructions and proofs of existing protocols becoming the bread and butter of academic cryptography. Previously unsolved problems could be solved, with the provably secure optimal padding for RSA being put forward for standardization [32]. Going even further, combinations of protocols could even be proven secure using proof techniques such as universal composabilty, where one could define the functionality in an "ideal world" and prove that a "real world" instantiation fulfilled certain security and privacy properties [43]. Work on indistinguishability obfuscation held the promise of being able to construct all the diverse branches of cryptography with the addition of "one-way" functions [44]. Large-scale work began systematizing a formal treatment of cryptography based in complexity theory [45]. For readers wishing to learn cryptography, textbooks on cryptography based on precise definitions with proofs are available [46]. The hope was that the design of cryptographic protocols could, by building on formally provable security, transform cryptography from an *informal art* to a branch of mathematics on par with any other [37].

Formalizing the adversary via logic

The nascent field of security engineering formed to analyze not just cryptography, but cryptography as part of the construction of larger systems of information organization. The revolution of public key cryptography allowed new decentralized models of access control, which became exceptionally important due to the increased adoption of the internet. Unlike previous access-control systems such as the Biba model, which assumed a finite number of secret keepers [21], the internet is an open world containing a vast number of different kinds of adversaries and secret keepers. In this new environment, a system needs to distinguish secret keepers from possible adversaries. This insight gave rise to the field of *authentication protocols*, where an agent could prove using cryptography that they are a legitimate secret keeper to a larger information organization.

The first use-cases were authentication between agents, each with a symmetric key on a trust server, in the face of communicating in the presence of an adversary attempting breaches of integrity and confidentiality, as well as "replay" attacks, where messages were held back or forwarded by the adversary. A host of strangely named home-brewed protocols, such as the Wide-Mouthed Frog and Yahalom protocols, were shown to be insecure upon closer inspection [47]. This led to the more rigorous Needham–Schroeder protocol for key distribution [47], which in turn became the basis for Kerberos, the first widely used authentication system on the internet [48]. Still, it was not clear if even the Needham–Schroeder protocol was secure. A number of attacks led to features being added in an *ad hoc* manner to the protocol, like "timestamps" to prevent replay attacks [49]. To address these issues, a revolution in formalization took place in the study of the security of these authentication protocols, strangely in parallel to the formalization of cryptography as it was led by a separate research community [50]. As opposed to the formalization of cryptography, where "cryptographic operations are seen as functions on strings of bits" and their security properties are "defined in terms of the probability and computational complexity of successful attacks," in the logic-based paradigm used by access control systems, cryptographic operations and their security properties "are seen as functions on a space of symbolic (formal) expressions" [50].

The general approach was that the communication of secrets could be studied in terms of the beliefs about the messages by both secret keepers and adversaries. Prior to the invention of cryptography, this was formalized in terms of epistemic logic, whose roots were in philosophy [51]. In a logic-based analysis, agents are abstract machines that can send and receive messages. In contrast to the investigation of the correctness of cryptosystems, cryptographic operations are assumed to be perfectly correct. However, the adversary is characterized in a completely different manner: Rather than a polynomial–time Turing machine, under the Dolev–Yao symbolic model the adversary is extremely powerful and "carries the message" over a hostile network, where the adversary can intercept, alter, and generate any message [52].

The first influential logic for authentication protocols, Burrows–Abadi–Needham (BAN) logic, applied an epistemic logic-based approach to existing authentication protocols, showing that the Needham–Schroeder was "secure" [53]. However, BAN logic lacked a standard model-theoretic semantics of possible universes, and protocols proven secure by BAN logic in theory were found to be actually insecure in practice due to the level of abstraction and limits in the logic itself [54]. In the wake of these deep-seated problems of BAN logic, researchers created a panoply of new logics: some with soundness properties [55], others based on the pi-calculus [56], still others on strand spaces [57], and well-defined multiagent epistemic logics [58]. The eventual victory of finding a suitable and more perfect logic appeared to be assured, and the unification of logic-based

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approaches with computational complexity was valiantly attempted for authentication protocols [50] by taming the powerful Dolev–Yao adversary in terms of the PPT adversary [59].

One advantage of being able to capture the philosophical notion of secrecy in standard logics is that protocols could then in theory be amendable to automated theorem-proving, where a machine rather than a human cryptographer generates the proofs using an automatic model-checker [60]. The core idea was that a run of the protocol, and all attacks by an adversary against it, could be conceptualized as moves in a space of all possible states of the protocol. A human cryptographer, with finite attention, was only likely to explore a relatively small subset of the space and thus only able to notice a few possible attacks. Alternatively, a machine could explore much more of the state space automatically—if not the entire state space—and find otherwise unknown attacks that would be too bizarre for a mere mortal to easily imagine without machinic help. There were of course computational limits in practice: These automated proof-systems were typically sound but not complete, and also often required interaction with a human to guide the heuristics used by the model-checker. Generally a human had to describe the model of the protocol and its security properties using whatever formal language was supported by the theorem prover before putting its automation to work. The early results were promising: A new attack was found on the venerable Needham–Schroeder key exchange protocol 17 years after the protocol was published [61], and this attack could be formalized using an automated theorem-prover based on the well-known Communicating Sequential Processes formalism [54]. The hope was that while Needham–Schroeder was a tiny protocol composed of only three lines of code, automated methods could be used to prove the security of very complex protocols like Transport Layer Security (TLS) that were based on thousands of lines of code. With the use of the higher-order logic-based proof-prover Isabelle [62], the very complex protocol TLS was also proven to be secure [63]. As a result, the triumph of formalism in cryptography seemed complete, and a bright future of cryptography—and its wider application in complex security protocols—seemed to be inexorable.

The question of what it means for a cryptosystem to be secure against an adversary is a philosophical question embodied as a technical question. First, if the larger philosophical notion of secure communication could be reduced to the sending of an encrypted message between secret keepers using a cryptosystem such that a known-class of adversaries could not decrypt the message, then in principle security could be reduced to a matter of mathematical proof. So while Shannon proved no adversary can break an information-theoretic and perfectly secure cryptosystem, where the key is never reused, in practice, these new results proved breaking a cryptosystem with key reusage is practically impossible if the adversary has a bounded amount of time and computational power. The initial application of computational complexity theory and logic-based formal verification made huge strides in solving some of these difficult questions for both cryptosystems and their usage in the authentication logics for access control systems. However, the philosophical question of whether or not there were fundamental limits to formal abstraction would show that questions of metaphysics and ontology had consequences for preserving secrets against adversaries.

The limits of formalization in cryptography

All was not well for formalism; systems had to be implemented by working code to be usable by secret keepers, but the capabilities of adversaries were not limited by any formal model. First, a number of protocols that were proven to be cryptographically secure in theory were shown to be insecure when implemented in practice [64]. The famous provably secure RSA padding proof was shown to have an error in the proof [65]. Worse, there was a brutal attack on RSA, which took advantage of TLS servers telling clients about improper padding, which could then be used to decrypt their encrypted traffic [36], despite the core of TLS being shown to be formally secure [63]. There was further worry that if protocols were proven secure under the Random Oracle assumption, they were not actually secure in practice as a hash function could be a poor approximation of a random oracle. After all, a random oracle is not even physically realizable by a finite computer, as it would require a description of infinite length; its underlying idea is that given infinite possible inputs, each input produces a random output independent from every other possible random output [66].

Despite the considerable success of formalization, the issue of concrete parameters also was still more guess-work than a proper science. For elements of cryptography like keys, bigger was always better: Yet, exactly what key size was reasonable to resist an adversary like the NSA with supercomputing resources? The move away from RSA to more efficient elliptic curve cryptography (ECC) exemplified concerns as it was not even clear which curve to use. Despite formal proofs, in practice security was based more on an oral tradition than a science as "the decision about what kind of curve to use in ECC is a subjective one based on the user's best guess about future vulnerabilities" [67]. The debate over the actual properties delivered by provable security came to a head when the inventor of elliptic curve cryptography [68], Neal Koblitz published an infamous paper with Alfred Menezes called *Another Look at Provable Security* that claimed "One has to wonder how many 'proofs' of security are ever read carefully with a critical eye" [69]. As they note, Mihir Bellare and Phil Rogaway's proof of optimal padding for RSA "went essentially unexamined for 7 years," so "one cannot help asking whether the lengthy and often poorly written "proofs" of less famous security claims are ever read carefully by anyone" [69]. Although one would hope a computer could verify these proofs, automated theorem provers effectively reduced the security of a protocol to the underlying—and perhaps flawed—model-checking software, which could be sound, but nonetheless was not complete and could not handle the complex assumptions of most proofs. The epistemological question of whether these computer-generated proofs could even be checked and trusted could not be banished [70]. Automated theorem provers produced proofs that were typically so complex and long that it would be impractical for any human to bother to check their correctness [71]. If the cryptographic research community did not regularly check important and concise proofs produced by its own members, what were the chances of their checking the tedious proofs generated by machines?

Despite building a large-case against the nature of reductionist security proofs in cryptography being error-prone and even circular [72], most academic cryptographers believed that abandoning security definitions and cryptographic proofs would be a step backwards. Otherwise, cryptography would be based on subjective intuition, and so prevented from being based on rigorous and scientific grounds [73]. This debate over the status of proofs within cryptography had a philosophical parallel in an earlier debate between the formalist and intuitionist approaches to the status of proofs in the foundations of mathematics. This philosophical debate arose out of the failure of Russell and Whitehead to construct the foundations of mathematics on logic [74]. For the formalists, what was important was that a set of axioms created a system of objects, and for each new object there can be a reduction in a consistent manner to the underlying model

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with no ontological commitment [75]. In contrast, intuitionism defined the origin of truth in mathematics in human experience, for it had "no other source than an intuition, which places its concepts and inferences before our eyes as immediately clear" [76]. For intuitionists, proofs force mathematicians to construct valid objects while avoiding constructions that make use of metaphysical presuppositions that could not be constructed by an effective algorithm grounded in finite human experience, such as the law of the excluded middle. Intuitionism thus offers justification via construction in place of a Platonic notion of the human-independent truth of mathematical objects [77]. For intuitionism, the disturbing feeling remains that the validity of a statement in a proof is then dependent on the cognitive faculties and subjective state of the individual mathematician, which inevitably leads us down the path of solipsism, where—per Feyerabend—"anything goes" [78]. In a world of adversarial behavior where consequential secrets are at stake, this nonchalant approach to security is an unacceptable path for most members of the cryptographic research community, and certainly the general public.

The question of the status of proofs can thus be restated in terms of the reality of mathematical objects. Are mathematical proofs real objects in terms of metaphysics? It would seem that mathematical objects could be as ontologically real as the rest of our world, or at least real structural regularities in our world [79]. While it appears feasible that one could claim that a perfect circle is some sort of Platonic object about which a mathematician could have *a priori* knowledge, it seems absurd to claim that Needham–Schroeder key agreement or RSA padding schemes have some sort of necessary ontological existence that precedes their invention. As noted by Rogaway [80], cryptographic programs are literally constructed by human engineers and cryptographers. Therefore, some variant of social constructivism over the nature of mathematical objects used to study cryptography would naturally fit the work of the cryptographic research community, which could then vary in opinion over the metaphysical status of these objects and their proofs [81].

However, there are dangers with replacing the objective reality with social consensus: "The cost is that "objectivity" is reinterpreted as "social," and although it still refers to something external to the individual it is no longer external to humankind" [81]. As per Lakatos, proofs are then constructed by all-too-human scientific communities who, while adversarial over the details, are united by their common desire for knowledge about their subject matter [82]. In cryptography, there is no common humankind, or even a unified research community; the common understanding of the social as put forward in the philosophy of science is a cooperative scientific community, but in cryptography humans are divided into secret keepers and adversaries. Thus, for a system to be secure against an adversary, it requires more than social consensus: The system must hold up its properties against full-blooded adversaries actively trying to undermine and find flaws in the work of the community of cryptographic researchers. More suitable is the approach of Daniel Dennett, who argues that mathematics, science, and even engineering can discover "real patterns" that are causally efficacious in the world, and so can be helpful in accomplishing evolutionary goals while avoiding Platonism and more stringent versions of scientific realism [83]. A cryptosystem may be socially constructed, but it may nonetheless discover and exploit a real pattern in the world; it must work in detecting and creating ontological regularities in the world that are independent of agents, whether or not they are secret keepers or adversaries. Whether or not something is secure then is a matter of its survival in the real world against adversaries, rather than against attacks made by the scientific community within an austere and pristine formal framework. Rather than being an arbitrator of the status of security, the cryptographic research community can be thought of as a testing ground—or, better yet, a collective laboratory—for new primitives and protocols.

Proofs thus provide an important role as "intuition-pumps" of the research community in finding weaknesses in security *before* a new cryptographic construction is created and in *repairing* already-existing cryptography in the wild [84]. The proofs produced by mathematicians—and accepted by the larger mathematical community—are often quite informal if not rigorous [85], and would require further transformations to count as a proof in a formalized logical system [86]. While many proofs are indeed informal and intuitive in cryptography, the historical tendency has been to move in the other direction by constructing increasingly rigorous logical proofs, including the aid of automatic theorem provers to detect unknown adversarial behavior. The very reliance on computing machinery to make proofs further undermines the claim that these proofs are a sort of Platonic *a priori* knowledge, rather than being quasi-empirical discoveries [70]. This positions cryptography as a discipline more akin to engineering or science than a relative of pure mathematics.

The received view is that engineering emanates historically from the application of science [87]. With cryptography and computer science as a whole, it is more often than not the other way around: Concrete engineering practices precede the slow work of scientific systematization in cryptography. This does not mean that the objects created by engineering are not real, but precisely the opposite: cryptography is a kind of *philosophical engineering* that is creating new ontologies with novel properties [88]. The new ontologies embodied by cryptographic primitives and protocols, whose security and weaknesses outstrip the intended design of their inventors, demand that mathematical and empirical work be done to determine the exact properties of these engineered artifacts in the wild. This is precisely because these embodied programs function in the real world while also being designed in the abstract with only a partial model of the world in the minds of their creators [89]. So cryptography can be considered as both designing new objects via engineering based on idealized models and making scientific, empirical discoveries about the operations—and security—of these objects in the real world.

The model is only useful in clarifying aspects of the entire system and so abstracts away from the rich details of their underlying subject matter. This is not only an issue for cryptography, as all formal models abstract from the real world. Even Hilbert and Bernays, the founders of the formalism, recognized that the work of mathematicians operates "predominately with theories which do not reproduce the actual state of affairs completely, but represent a *simplifying idealization* of the state of affairs and have their meaning therein" [75]. The problem with relying on purely formal models for security properties is that, as Cantwell Smith puts is: "Mathematics is unique, with respect to models, because (at least to a first level of approximation) its subject matter is the world of models and abstract structures, and therefore the model–world relationship is relatively unproblematic" [5].

The subject matter of formal models in cryptography, therefore, is not mathematics—as would be the case in meta-mathematics—but rather an engineered system facing an adversary. The limits of formalization in cryptography are due to the model *of* a system being mistaken for the system itself as embedded in the world, which includes the social behavior of secret keepers. Likewise, a model *of* an adversary cannot be mistaken for the actual adversary, which could possess attacks unconstrained by any model. The root of the problem was that formal definitions that undergirded the entire field of provable security: the definitions were idealizations of reality, which failed to take into account important aspects of the implementa-

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tion, and it was precisely the aspects left out of the idealization that were often crucial for a successful adversarial attack. For example, semantic security assumed that the length of the encrypted message was public, as in theory a message may be of unbounded length [34]. In practice messages in the real-world did have varying lengths, and the behavior of encryption algorithms in the face of varying lengths led to real-world attacks like CRIME and BREACH on TLS [90].

The philosophical question of the limits of formalism is of prime importance to cryptography. Debates within cryptography echo debates earlier in the philosophy of mathematics, ranging from debates over whether mathematics provides privileged access to the nature to reality of whether or not the metaphysical complexity of the world outstrips any formal model of the world [5]. If there are indeed fundamental limits to correctness in computers, then even automated theorem provers would never truly be able to prove that security properties would hold in the real world [89]. As Ross Anderson notes in his seminal paper *Why Crypto-systems Fail*, the engineers who construct these systems and the cryptographic community that attempts to prove their security properties "have suffered from a lack of information about how their products fail in practice, as opposed to how they might fail in theory," as an adversary would typically not reveal that they had broken a cryptosystem [91]. The reason for cryptography's lack of the virtuous feedback cycle of other scientific disciplines and engineering research fields was simple: secret keepers that reveal their secrets due to broken cryptosystems may also be loathe to reveal their failure or not even know that their secret has been revealed.

The adversary revealed

The limits of formalization are revealed when the capabilities of an adversary are not captured by a formal model. While the notion of an adversary had been formalized with increasingly sophisticated abstractions, the details of how actual nation-state level adversaries worked remained speculation. With the leaks by Edward Snowden, the unthinkable happened: the raw secret techniques of the most powerful adversary yet imagined were revealed to the general public. This guiding example of the limits of formalization was an event of historic proportions to the field of cryptography as, precisely because the NSA was the most powerful adversary imaginable in terms of cryptography, academics had long-used the NSA as a sort of mythical adversary in their research papers. If a technique could defeat the power of the NSA, it was surmized to be able to withstand the attacks of any lesser adversary. Before Snowden's leaks, although the academic cryptographic research community knew that the NSA had long suppressed the release of public key cryptography, their precise capabilities as an adversary could only be guessed [23]. For example, the discovery of a large secret server farm under construction in Utah was surmized to be a cryptanalytic facility aimed at the "brute-forcing" of private key material, at least judging by its massive size and its energy consumption [92]. The Snowden leaks finally offered a glimpse into what a large-scale adversary with thousands of employed secret keepers and more than a billion dollars in budget was capable of. Although the number of secrets leaked by Snowden are clearly far from exhaustive and rapidly dating, we will touch upon a number of findings that revealed blindspots in the prior philosophical framing of cryptography [93].

The first surprise was that defeating cryptography via breaking a cryptosystem was not the primary methodology employed by a real-world adversary to reveal secrets. Although the NSA was active in cryptanalysis, there was no evidence of their having hidden revolutionary cryptographic techniques to break existing cryptosystems: The polynomial adversary did seem to be an accurate assessment of the NSA's cryptanalytic capabilities, and even the NSA could not defeat modern cryptography. Years earlier it had been shown that the NSA had discovered and deployed differential cryptanalysis, a powerful mathematical technique capable of breaking early cryptosystems, before the technique had been discovered by the cryptographic academics [94]. Instead of cryptanalysis, the NSA was obsessed with obtaining secret key material via attacking parts of the system left out of standard information-theoretic model of cryptographic communication from Shannon onwards. Why would an adversary attack the encrypted messages when it could more easily attack the secret keepers themselves? Often key material was stored insecurely by the sender or receiver of a secret and could be extracted by attacking the secret keeper's device via malware (as shown by "Foxacid") [95]. Rather than attack the message, the communication channel itself could be subverted; such malware could even be deployed against the communication channel itself in order to access sensitive data, as shown by an attack the British Government Communications Headquarters (GCHQ) did against a Belgian telecommunications provider in order to obtain secret information from their nominal allies [95]. Implementation details left out of abstract models of protocols were ripe for attack: The NSA had a particular focus on subverting the TLS protocol used to secure access to websites. One technique, called "Quantum Insertion," was simply injecting fake packets into an otherwise seemingly secure TLS connection in order to redirect an agent toward a malicious website of the attacker's choosing [95]. Lastly, the point-to-point model of communication used since Shannon did not take into account massive trusted third-party platforms that mediated most communication channels. Via the PRISM programme, the NSA was revealed to have access to Microsoft, Google, Facebook, Apple, and other large centralized platforms, including access to the plaintext of emails and web data.

The second finding was that the simplest primitives used in the reductions for security proofs could be attacked far more easily than the protocols itself. Via Project Bullrun, the NSA had a budget of 250 million dollars to subvert cryptographic standards [96]. This was no surprise, as standards like DSA for digital signatures, created by NIST but designed by the NSA, were clearly crippled to the extent that DSA was both inefficient and could only be used for authentication, not confidentiality [23]. On the one hand, major standards by NIST long-rumored to be weakened by the NSA—such as the symmetric key encryption of AES—were shown not to be broken by the NSA in the Snowden leaks. On the other hand, Snowden confirmed some long-standing suspicions about the NSA subversion of the NIST Dual_EC_DRBG (Dual Elliptic Curve Deterministic Random Bit Generator) standard [96], which was also internationalized by the International Standards Organization [97]. Cryptosystems assume that the keys are generated via random number generators, but a subtle "backdoor" was put in the random number generation so that even though its output appeared random, the possessor of the backdoor information could predict the generated keys. This insecure random number generation meant that TLS connections using it could be compromised. The NSA even worked with vendors—such as the RSA company, although without the knowledge of Ron Rivest and his coinventors—to ensure placement of the insecure random number generator in high-value corporate infrastructures [98]. This shocked the cryptographic research community, and Rivest himself was surprised at the poor judgment of his own company. Furthermore, this strategy of having deliberately weakened standards or products with backdoors could easily be applied to other crypto-

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graphic system, which the NSA would then help place in commercial products. Of course, this strategy to gathering secret information is technically questionable: as soon as such a backdoor was discovered, it would be usable by any other adversary and in no way limited to the party that designed the backdoor. This seems to have been the case with the NSA backdoor in Dual_EC_DRBG: while it let the NSA decrypt the internet traffic of Juniper routers, after the technique went public it appears that Chinese intelligence also took advantage of the backdoor [99].

The last finding of the Snowden leaks was that the patterns of messages given by secret keepers was just as important, if not more so, to the adversary than the encrypted message itself. While the most fundamental problem in cryptography is the protection of secret information being sent from a sender to a receiver, another even more difficult problem is how to send a message without revealing the identity of the sender and receiver to an adversary. In other words, how can one hide not just the *data* of the encrypted message, but also the *metadata* of who is communicating with whom? Techniques to do this would unlink the sender and the receiver of a message so that they can communicate anonymously [100]. In short, the pattern of communication revealed the information organization of the secret keepers to the adversary, and keeping the organization's structure secret was itself more important in many cases than the secrets themselves. Take for example the organizers of a protest in a country under authoritarian rule, who must keep the information about the time and location of the protest hidden, but also their own identities to escape arrest. Even if the message was encrypted, often the encrypted message has metadata that is itself plaintext on the level of the protocol implementation, such as the name of the website being visited using TLS. The monitoring of messages, including encrypted messages, for patterns in communication by an adversary is called *traffic analysis*, and as Diffie and Landau put it: "Traffic analysis, not cryptanalysis, is the backbone of communications intelligence" [15]. The Snowden leaks showed that the NSA was a master of traffic analysis: the slogan of the NSA was "Collect it all, know it all, exploit it all" [93]. With millions upon millions of messages collected everyday across almost every country, the NSA could do large-scale data analysis and visualization of social networks via programs like "Boundless Informant" and personalized monitoring in real-time via "Keyscore-X" [101]. VPNs and anonymous communication networks such as Tor deliver messages from sender to receiver in a first-in, first-out manner through a series of decentralized relays [102]. Existing real-world techniques were insufficient: VPNs and even Tor were vulnerable to traffic analysis by a powerful adversary such as the NSA that could capture and record every message, although deanonymizing Tor users took considerable manual work by the NSA due to its decentralized nature [103].

What sort of research problems were provoked by the NSA that went beyond the prior formal frameworks? The Snowden leaks shocked not only the general public, but many members of the cryptographic research community to their core. One of the founders of provable security, Phil Rogaway, wrote an eloquent call to arms, entitled *The Moral Character of Cryptographic Work*, where he said: "As computer scientists and cryptographers, we are twice culpable when it comes to mass surveillance: computer science created the technologies that underlie our communications infrastructure, and that are now turning it into an apparatus for surveillance and control; while cryptography contains within it the underused potential to help redirect this tragic turn" [104]. Having the adversary revealed caused a transformation of the paradigm of the field of cryptography, at least for researchers like Rogaway [105]. Although the majority of the cryptographic research community continued with little thought to real-world implementation or concerns, a portion of the cryptographic research community did work to resolve the most egregious attacks brought to light by Snowden. New protocols like Signal were created for secure messaging that featured new properties like *forward secrecy*, where the compromise of a key did not allow the adversary to decrypt all past and even future messages [106]. The ancient TLS protocol was finally properly analyzed, and new techniques in formal verification allowed the model-checking software to go beyond the mere analysis of a model to actually generate code that was verified to embody the model [107]. Focus on key storage in hardware that was difficult to extract via secure enclaves also became a major focus of work [108], as did the logging of key material to prevent forgeries (including for TLS) [109] and the decentralization of information away from large platforms, leading eventually to interest in blockchain-based systems to preserve the integrity of information against adversarial and centralized systems [110].

Due to the Snowden leaks, the field of provable security had to revisit its formal assumptions: the NSA subversion of cryptographic primitives and basic protocols was a thorny problem to address. Suspicions naturally fell on the NIST-selected elliptic curves, whose precise choice of parameters were difficult to publicly verify, and which seemed less than ideal in terms of efficiency. Daniel Bernstein and Tanja Lange created an alternative with the peer-reviewed Curve25519 curve that was even more efficient than the standardized NIST curves, and it soon found its way into major standards like TLS [111]. Another research paradigm, called *malicious security*, had been explored earlier in terms of viruses and malware by a small group of cryptographers to determine if a protocol could be made secure even if the fundamental primitives to which proofs reduced security were subverted [112]. In the wake of the possibility of standardized primitives being subverted, this line of research could then be applied to the assumptions of modern cryptographic protocols [113].

Traditional cryptographic research did not focus on enabling anonymous communication, and what little research existed on anonymous communication was woefully underdeveloped [114]. Although the field of cryptography had been well-developed mathematically, the study of anonymity still seemed stuck at the prescientific stage of determining a basic terminology and methodology [100]. While work in epistemic logic can formalize anonymity, it has not been united with other models [115]. In contrast to traditional work in cryptography, work in anonymity consists of hiding metadata details, like timing and volume information of encrypted messages. However, widely used anonymous communication networks such as Tor did not protect against a *global passive adversary* that could watch and copy every single message on a network [102]. Prior to the Snowden leaks, this kind of adversary was considered unrealistic, but post-Snowden it became clear this kind of mass data collection was already possible. This led to renewed interest in mix networking, a technique invented by David Chaum in which encrypted messages were shuffled like a deck of cards in a series of decentralized servers, and so delivered to receivers in a different order than sent, effectively anonymizing the message [114]. After laying dormant for almost 30 years, mixnets like Nym were developed to prevent the traffic analysis of users, exemplifying the next step in protocol development beyond TLS [116].

The adversary may have been revealed, but the full extent of their powers is still unknown as the Snowden leaks gave only a glimpse of their cryptanalytic power at a particular point in time. A further paradigm shift awaits, as the invention of a real-world quantum computer, by implementing Shor's algorithm to exponentially parallelize factoring, might break the fundamental trap door assumptions that

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were built into almost all cryptography ranging from Diffie–Hellman key exchange to elliptic curve cryptography [117]. Such a quantum computer no longer seems distant, and could feasibly be created by the intelligence agencies of large nation-states within decades or less. Therefore, the cryptographic research community, working hand-in-hand with NIST, standardized a new series of postquantum replacements for public key cryptography based on new mathematical problems, such as the shortest vector problem that would resist a quantum computer [118]. Nonetheless, the question remains: can the NSA and NIST be trusted in the context of using much less-studied mathematical problems to create a new generation of cryptography?

The Snowden leaks pulled the mask off the NSA, showing that powerful global passive adversaries and active attacks on all aspects of communication are a realistic threat model. From a philosophical perspective, the Snowden leaks provide an example of how both our initial philosophical framework can analyze an adversary like the NSA, while demonstrating the limits of various abstract models, for example, in capturing the NSA's actual capabilities. Of course, being an adversary is a relative notion: For the NSA, Snowden himself was the adversary that leaked their secret techniques. Snowden's leaks also brought to popular attention a political and philosophical shift: cryptography was historically a weapon of defense used by large-scale information organizations of secret keepers such as governments, yet now tools such as big data analysis and the exploitation of security flaws were focused on individuals in the general population, even indiscriminately in the case of mass surveillance. The adversaries were also changing. On the one hand, individuals were increasingly empowered to guard their own secrets with cryptography due to the invention of public key cryptography. On the other hand, the attacks that were once considered only the province of the NSA, such as brute-force attacks and malware, increasingly became usable by not only other intelligence agencies, but also a host of non-state actors ranging from criminal networks to private organizations. The successful formalization of the adversary had effectively caused cryptography to ignore important aspects that fell outside this model: attacks on the communication channel and secret keepers themselves, the subversion of cryptographic primitives, and the statistical analysis of the social network of secret keepers. The kernel of a future research agenda had been revealed.

Conclusion: a research agenda

Philosophy has long been the handmaiden of science, yet in the case of cryptography the question of the adversary seems to reveal a lacuna at the heart of philosophy as well. One should not forget that the translation of "Satan" in Hebrew is adversary, and "programming Satan's computer"—where the very channel to carry communications can be malicious—is one of the most difficult tasks imaginable [119]. Politics has long been the realm of adversarial behavior, but now—as shown by the Snowden leaks and the increased ubiquity of attacks on digital communication of all sorts—the problem of communicating in the presence of adversaries is also a technical question. This technical question impacts not only governments and other information organizations, but nearly all of humanity as we increasingly become dependent on the Internet. And, as has been shown by our analysis of cryptography, any technology is built on a robust, if implicit, philosophical foundation that is worthy of study in its own right.

While cryptography has come quite a long way in defending communications, the purpose of a philosophy of cryptography should be 2-fold: first, to make philosophical insights legible to cryptographers; and second, insofar as information theory has served as the fundamental building block of the philosophy of cognitive science, artificial intelligence, and even the newly founded philosophy of information, there is no reason why philosophers should not apply their philosophical machinery to cryptography. Rather than view cryptography as a pristine, if not bizarre, branch of mathematics where philosophers have to be called in to tidy up the foundations, it would be better to use the insights provided by philosophy to point towards ways of expanding cryptography into a more general science of security.

In order to qualify as a science, abstraction must be tempered by empirical research in order to allow secure communication in reality, not just on pen and paper. So far the progress of cryptography has critically depended since Shannon on the advancement of abstraction, yet future progress may require—in order to not mistake the map for the territory per Korzybski—the *reverse* of abstraction: *reconciliation* with the underlying subject matter. Cantwell Smith defines reconciliation as "filling in the details" of any theory [5]. This could allow us to revisit our abstractions and choices to produce more accurate models that do justice to their underlying subject matter, whether in terms of predictions or even just explanations. We suggest the following four questions for a research agenda to reconcile the drive towards abstraction in cryptography with its embedding in the world, where each question is to be explored in more detail.

(1) Can we study dynamic social networks of secret communication modeled on the empirical behavior of secret keepers?
(2) By incorporating statistics, can new models of the adversary take into account more accurate data about their capabilities?
(3) Can cryptography incorporate interdisciplinary methods and insights that can capture the complexity of secure communication?
(4) Will philosophy let us make explicit and then reformulate the hidden assumptions of cryptographic communication, and *vice versa*?

First, can we study dynamic social networks of secret communication modeled on the social behavior of secret keepers? As initially explored by access control and authentication logics, the question of the secret is a question of information organization. Adversaries are not limited to technical means, they may actively be attempting to reveal secrets of their targets by intercepting the message, but also monitoring or interfering with—or even subverting—the behavior of secret keepers. Today, information organization is thought of as a social network rather than a simplified centralized organization. It is at this point, we should note the limits of Shannon's simplification of secure communication is insufficient: as demonstrated, albeit negatively, in spades by Snowden's insider leak of one of the world's most powerful organizations of secret keepers, information organization is a norm-governed social relationship between secret keepers and their secret that prevents this secret from escaping their agreed-upon bounds and so falling into the hands of the adversary. In the traditional model of Shannon, using conventions developed in the original treatment of RSA [29], there is only one channel of communication between two agents Alice and Bob, with a single malicious adversary—known as Eve or Mallory—who is attempting to maliciously disrupt or eavesdrop upon the communication, with some perhaps backup cast of trusted third parties or other infrastructure so that this model would hold by induction to any number of channels.

This is an admirable strategy at a relatively high-level of abstraction, but in order to further reconcile cryptography with the messy details of real-world attacks, the simplified structure of communica-

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tion should be extended to a heterogeneous techno–social network of communication between a large cast of agents, where these agents can have varying level of capabilities in terms of cryptosystems, key storage, and so on. The agents are given as vertices (nodes) and the channel as an edge (link) between them. The key would then be to trace the flow of secret information between these various agents. Whether or not a given network of secret keepers is successful in their keeping a secret requires them to conform to a policy, which is an implicit or explicit set of rules defining who is in the circle of secret keepers versus who is not, as well as their norm-governed behavior. Whether or not this key invariant of secrecy is preserved over various changes in the dynamics of network structure would then require either modeling or empirically inspecting the information flows in different situations [120]. Although leaks in information flow are self-evident in the case of the whistle-blower, there is also the case of accidents and the like, so that the behavior of the adversary may be properly considered a byzantine fault; in the real world it may not be possible to distinguish between operational failures and malicious behavior when a secret is leaked. By thinking security in terms of the flow of secrets over real-world networks, real (or worse case, synthetic) data can be attached to otherwise far too abstract cryptographic protocols and access control systems. This allows research in cryptography to go beyond simple maxims from philosophy and social science on adversarial behavior. Of course, this is often hampered by the difficulty in getting data about attacks, as most organizations are loathe to share potentially embarrassing data [121]. The collection of such data, when properly anonymized, should be considered a common goal for all parties to increase their own security by learning from the lessons of others.

Second, by incorporating real-world data, can new models of the adversary take into account more accurate data about their capabilities? One of the most pernicious problems facing cryptography is an accurate description of the adversary. While the abstractions such as the polynomial–time adversary and the Dolev–Yao adversary have been historically useful, they are far too abstract for constructing realistic adversarial models [34,52]. Adversaries are also not a simple universal enemy, but may be constituted by real-world networks between humans and machines, and they may not be operating alone but rather in coalition with other adversaries. Understanding these adversaries in excruciating detail will require both new kinds of logic and models that are informed by empirical data, which in turn will require an incorporation of statistical methods. For an example in the context of mix networking, statistical techniques like Monte–Carlo simulation allow us to define the probability of an local adversary linking a particular message to a sender or receiver even when the adversary has only a partial view of an entire network [122]. In terms of logic, more detailed logical models from distributed systems and the semantics of information theory, which can potentially line up with the behavior of these models as complex systems, have been underexploited in cryptography [123,124]. What will be needed is not less abstraction, but a greater variety of abstractions and mathematical tools to effectively deal with the real-world data of adversarial communication.

The political economy of the surveillance powers of an adversary should also be taken into account. For example, in the "Five Eyes" alliance as leaked by Snowden [125], the USA operates with alliance with Canada, the UK, and other Anglophone countries by sharing intelligence capabilities. Some parts of this coalition may have specialized capabilities, such as the tremendous computing facilities of the USA, while others like the UK's GCHQ may be capable of trapping undersea cables in Europe, a task that was outside of the rules of the USA at the time. Not all members of an adversarial coalition are equal; coalitions may also have different levels of access to capabilities, such as the ability to ask others for surveillance capabilities but not being able to autonomously deploy that capability: for example, Saudi Arabia will be forced to choose between using Chinese and American capabilities for surveillance, leading Saudi Arabia's infrastructure to be a large extent subject to a form of cybercolonialism [126]. However, other countries like Iran may chose to obtain their surveillance capabilities from the Russia–China nexus. These adversarial coalitions are not necessarily limited to nation-states, but may be adversarial coalitions within nation-states: for example, surveillance capabilities are typically not shared directly between the NSA and FBI in the USA, although the NSA needs to alert the FBI if they encounter information relevant to a domestic investigation. More commonly, coalitions may be between superempowered individuals and loose social networks with ever blurring borders, such as the hacking campaigns of Anonymous [127]. Adversarial coalitions between nation-states and individuals allow them to combine forces (as often is seen in Russian hacking) and state-level capabilities may target individuals and other nonstate actors, as exemplified by North Korean malicious malware to obtain financial resources for their nuclear program [128]. These coalitions may have vastly different access to inside information, cryptanalytic abilities, computational power, and more. Therefore, in order to turn cybersecurity from an art to an empirically driven science, the first commandment should be to know the adversary: to create as an effective and complete model as possible of the real-world capabilities of the adversary.

Third, can cryptography incorporate interdisciplinary methods and insights that can capture the complexity of secure communication? Information flows between dynamic networks lend themselves to the theory of complex systems [129]. With suitable fine-grained descriptions of the communication channels between the secret keepers and the adversaries, invariants may be observed and predictions made while retaining fidelity to the underlying the subject matter. For any given description of a dynamic communication network, the behavior and evolution of this network can be described over time at various scales, with possible trajectories outlined due to adversarial attacks [120]. As is characteristic of complex systems, extreme attacks by the adversary may result in entire phase shifts in the structure of the communication network. Within this matrix of interdisciplinary work around complex adaptive systems, the social embedding of cryptography can be fruitfully studied [130]. In the emerging fields of network dynamics and nonparametric statistics—as most of the statistics will be done over social structures that infamously exhibit "power law" distributions—progress in modeling may be made, and research could profit from, similar work in studying the dynamics of information flow in related fields like epidemics [121]. With enough data and rich-enough models, this work could allow data-driven simulations to reveal possible invariants in terms of the preservation of secrecy. Yet, simulation-driven work has its own philosophical quandaries, particularly if the models reach a level of such complexity that their results are epistemologically opaque: a scientific invariant may be evident and have predictive power, but the reasons for it may not be known [131]. Complex models should always then be simplified and pruned, and then refined to not become too heavyweight. The theoretical apparatus of cryptography will naturally need to expand by incorporating rigorous approaches from economics, psychology, and philosophy that embrace complexity. This would transform cryptography into a general-purpose science of adversarial behavior, which could be called a science of information security, otherwise known as cybersecurity [132].

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Lastly, will philosophy let us make explicit and then reformulate the hidden assumptions of cryptographic communication, and vice versa? We should go beyond the simplified network models and calls for more data and complexity, to inspect the received wisdom of cryptography via the universal solvent of philosophy. On a large scale, our entire modern world is constructed from networks of communication in an adversarial environment. Even our own selves may be constructed in such a dynamic manner with our environment, and the simplified models that map humans to nodes and their communication to links in networks will steadily be overturned by careful philosophical analysis, as sharp boundaries between agents and their environment find themselves increasingly difficult to hold. This should lead to the questioning of Shannon's foundational model of secrets being sent over channels between discrete agents, as explored in our own preliminary philosophical investigation into cryptography. As shown by the Extended Mind Hypothesis, intelligence is distributed between the cognizing agent and the environment, and so agents may have fluid boundaries that dynamically integrate parts of their environment, including their digital environment [133]. Still, rarely have attacks on our increasingly vital digital infrastructure been taken as seriously as physical attacks [134], despite our cognitive infrastructure and our social institutions being increasingly digital. There is also much to be learned by philosophy from cryptography. Indeed, all sorts of questions from ontology to epistemology can be recast in explicit adversarial modes. It was such adversarial debate that gave birth to philosophy via Socrates, and philosophy should revisit the adversary as its foundational principle.

The philosophy of cryptography is the philosophical analysis of increasingly rigorous abstract models of adversaries, where this abstraction came at the cost of blinding cryptography to the very real capabilities of the adversary that were outside these models. The Snowden leaks brought some of the adversarial capabilities of the NSA to popular attention, and today there are many adversaries with diverse, if not even more powerful, adversarial powers due to the progress of technology. For example, the utilization of the mass collection of surveillance-based data by artificial intelligence is already underway, which will lead to increasingly powerful adversaries that can try to predict, control, and manipulate humans. A philosophical analysis of cryptography can point to a research agenda to reconcile these abstractions with the scientific, epistemological, and even ontological concerns brought to the forefront by the metaphysical richness of the world. It was a historic accident that the internet was designed in a trusted academic environment, and yet today we are not only universally connected, but universally vulnerable to adversaries at a global scale. Historically, the presence of adversaries has been a driver of biological evolution and social history, and so it is through secure communication and the organization of information that we as a species are likely to steer through global crisis. A world without adversaries would be utopian, but also static. A world inhabited by adversaries is not only a dangerous world, but one of dynamism and growth. In this regard, philosophy should be grateful for the adversary.

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Author's contribution

Harry Halpin (Conceptualization, Investigation, Writing – original draft, Writing – review & editing).

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